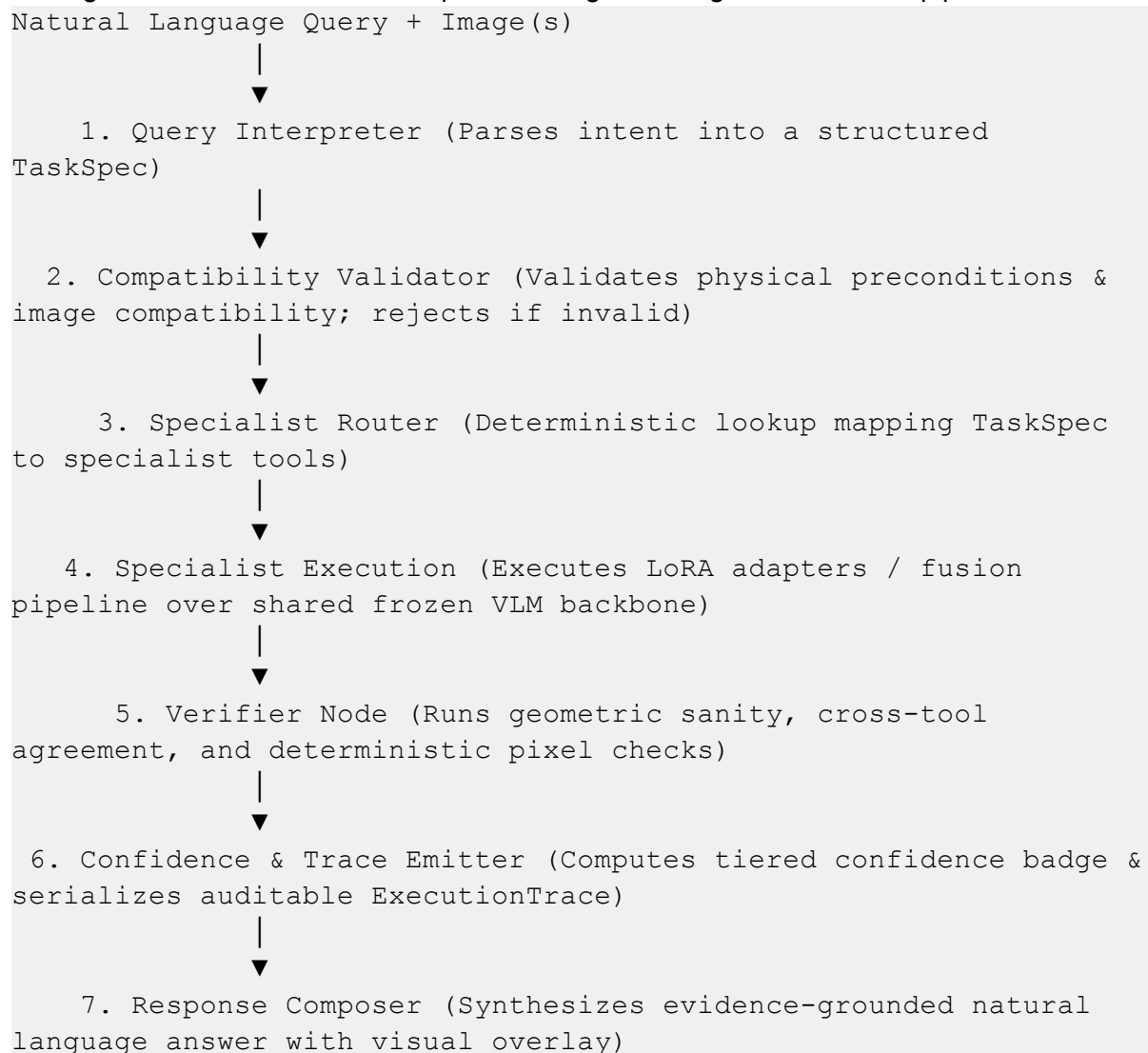


In **SatQuery AI, Agentic Model and Tool Orchestration** refers to the system's ability to act as an intelligent, query-driven controller that interprets natural language, validates input satellite imagery, selects and sequences specialized remote-sensing tools, and combines their outputs into auditable, evidence-grounded responses 1-4.

Rather than relying on a single unconstrained Large Vision-Language Model (VLM) or a fragile free-form ReAct loop—which frequently suffers from tool hallucination and argument formatting errors 5-8—SatQuery AI uses a **typed, schema-validated execution graph** governed by a strict **"validate before you execute"** architectural pattern 4, 9, 10.

Key Stages of the Orchestration Pipeline

The agentic controller executes requests through a 7-stage, deterministic pipeline 10-12:



1. Query Interpreter

Translates the user's free-text natural-language query and image metadata into a structured TaskSpec JSON object using a small, text-only instruct LLM in structured function-calling mode 3, 10, 13. If intent confidence is low or ambiguous, it triggers a clarifying question rather than guessing 13, 14.

2. Compatibility Validator

A deterministic node (operating without an LLM) that checks the requested TaskSpec against machine-readable physical preconditions before any specialist model runs 4, 15, 16. It validates:

- **Image count & formats** (GeoTIFF, TIFF, PNG, JPEG) 16-18.
- **Sensor modalities** (Optical, Multispectral, SAR) 15, 18.
- **Geospatial & Temporal properties** (Coordinate Reference System CRS, footprint overlap percentage via IoU, and acquisition dates) 15, 18, 19.

If a query is physically unsatisfiable (e.g., requesting a bi-temporal change analysis on images covering non-overlapping regions), the validator issues a hard **ValidationRejection** with a clear, human-readable explanation, halting execution before compute resources are wasted 4, 7, 9, 15, 19, 20.

3. Specialist Router

Performs a deterministic, non-LLM lookup from the validated TaskSpec to an ordered list of specialist tools (e.g., single-image VQA, text-guided grounding, bi-temporal change detection, or optical-SAR fusion) 3, 10, 21, 22.

4. Specialist Execution Layer

Invokes task-specific specialized tools 3, 22. Rather than running multiple monolithic models, SatQuery AI uses a **shared frozen vision-language backbone** and hot-swaps lightweight, task-specific Low-Rank Adaptation (**LoRA**) adapters (for VQA/captioning, grounding, and change analysis) or runs a structured 2-stage optical-SAR fusion pipeline 23-28.

5. Verifier Node

Cross-checks the collected evidence before presenting results to the user 23, 29, 30:

- **Geometric Sanity:** Verifies bounding boxes lie within normalized image bounds [31] and change masks do not exceed co-registered footprint overlap 29, 30, 32.
- **Cross-Tool Agreement:** Checks for consistency between independent specialists 29, 30, 32.
- **Deterministic Quantity Check:** For counting or ratio queries, computes deterministic pixel counts from change masks and compares them against the LLM's verbalized claims 30, 33, 34.

6. Confidence & Trace Emitter

Computes a **High**, **Medium**, or **Low** confidence badge using decoupled perception and reasoning signals 29, 32, 35-37. It applies structural discounts for known weak points (like counting or small-change queries) or hard downgrades if the Verifier detects geometric/quantity discrepancies 32, 33, 35-37. Simultaneously, it serializes a complete, auditable **ExecutionTrace** detailing selected tasks, model IDs, parameters, timing, and outputs 3, 29, 31, 38, 39.

7. Response Composer

The controller synthesizes the collected evidence ledger into a natural-language response, explicitly citing which sensor modality or specialist tool provided each specific claim 2, 3, 12, 25, 29.

Why This Design Matters

Feature, Generic ReAct / LLM Wrapper, SatQuery AI Agentic Orchestration

Tool Selection, "Free-form LLM decisions prone to tool hallucination 5, 7, 40", "Typed graph with deterministic schema routing 4, 7, 9, 22"

Input Handling, Passes raw images directly to tools regardless of fit 7, "Compatibility Validator structurally rejects unsatisfiable inputs upfront 4, 7, 9, 15"

Verification, "Unchecked generative output 7, 40", "Verifier Node enforces geometric bounds and pixel-level cross-checks 23, 29, 30, 32"

Auditability, Opaque internal reasoning or chat logs 40, "Serialized ExecutionTrace logging tasks, tools, parameters, and step timings 3, 31, 38, 39"

 *Would you like to explore the specific precondition parameters checked by the Compatibility Validator, or examine how the Verifier Node handles quantity discrepancies during change detection?*